

ECE 590.17: Lecture 6 – Learning Factor Graph Parameters

Graphical Models and Inference for Machine Learning

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History: Written by Henry Pfister (2026).

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6.1 Outline

In this lecture we focus on *parameter learning* for graphical models when the *graphical structure is known*. That is, we assume we are given a factor graph (or a Bayesian network DAG, or a Markov random field graph), and we want to learn the *local factor functions* from data.

What students should already know. We assume you are familiar with: (i) the definition of factor graphs and how a global function factors as a product of local functions; (ii) the definition of a probability distribution from an unnormalized factorization $\mu(x) = \frac{1}{Z} \prod_a f_a(x_{\partial a})$; (iii) belief propagation (BP) on trees for computing marginals and partition functions; (iv) basic probability and the phrase “maximum likelihood,” but not necessarily how to compute maximum likelihood estimates in practice.

What we will do. We will treat three increasingly challenging settings:

1. **Directed models (Bayesian networks), fully observed data.** We will derive the maximum likelihood estimator (MLE) for discrete conditional probability tables (CPTs). This case is conceptually important and computationally easy.
2. **Undirected / factor graph models, fully observed data.** We will see that MLE becomes an optimization problem involving the partition function. The gradient takes a simple

“data moment minus model moment” form, but computing the model moment requires inference. On trees, BP gives exact inference, so learning is efficient. On loopy graphs, we use approximations (loopy BP, pseudo-likelihood, sampling, etc.).

3. **Missing data / latent variables: EM using BP.** When some variables are hidden, the log-likelihood contains a log-sum over hidden configurations. Expectation–Maximization (EM) converts this into alternating steps: the E-step runs inference (BP) to compute expected sufficient statistics, and the M-step updates parameters (often in closed form for discrete factors).

One unifying theme. Learning in graphical models often reduces to repeated inference. So: *inference is the inner loop, learning is the outer loop.*

6.1.1 Notation and setup

Let $X = (X_1, \dots, X_n)$ be discrete random variables with finite alphabets $\mathcal{X}_i$ and product alphabet $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_n$. The goal is to learn a model from an observed i.i.d. dataset

$$\mathcal{D} = \{x^{(1)}, \dots, x^{(N)}\}, \quad x^{(\ell)} \in \mathcal{X}.$$

For any subvector X_S , we write the empirical probability under $\mathcal{D}$ of $x_S \in \mathcal{X}_S := \prod_{i \in S} \mathcal{X}_i$ by

$$\hat{p}_{X_S}^{\mathcal{D}}(x_S) := \frac{1}{N} \sum_{\ell=1}^N \mathbb{I}\{x_S^{(\ell)} = x_S\}.$$

Thus, we can write the empirical average of an arbitrary function $g : \mathcal{X} \rightarrow \mathbb{R}$ as

$$\hat{\mathbb{E}}_{\mathcal{D}}[g(X)] := \frac{1}{N} \sum_{\ell=1}^N g(x^{(\ell)}) = \sum_{x \in \mathcal{X}} \hat{p}_X^{\mathcal{D}}(x) g(x).$$

Definition 1 (Maximum likelihood estimation). Given a parametric family $\{p_X^\theta(x)\}_{\theta \in \Theta}$ of distributions on $\mathcal{X}$, the *log-likelihood of the dataset $\mathcal{D}$* is given by

$$\mathcal{L}(\theta) = \ln \prod_{\ell=1}^N p_X^\theta(x^{(\ell)}) = \sum_{\ell=1}^N \ln p_X^\theta(x^{(\ell)}).$$

A *maximum likelihood estimate* of θ satisfies

$$\hat{\theta}_{\text{ML}} \in \arg \max_{\theta} \mathcal{L}(\theta).$$

Why the logarithm? The logarithm turns products into sums and connects to cross-entropy / KL divergence: maximizing likelihood is equivalent to minimizing $D(\hat{p}_X^{\mathcal{D}} \| p_X^\theta)$:

$$\begin{aligned} \frac{1}{N} \mathcal{L}(\theta) &= \frac{1}{N} \sum_{\ell=1}^N \ln p_X^\theta(x^{(\ell)}) \\ &= \sum_{x \in \mathcal{X}} \hat{p}_X^{\mathcal{D}}(x) \ln p_X^\theta(x) \\ &= - \sum_{x \in \mathcal{X}} \hat{p}_X^{\mathcal{D}}(x) \ln \frac{1}{\hat{p}_X^{\mathcal{D}}(x)} - \sum_{x \in \mathcal{X}} \hat{p}_X^{\mathcal{D}}(x) \ln \frac{\hat{p}_X^{\mathcal{D}}(x)}{p_X^\theta(x)} \\ &= -H(\hat{p}_X^{\mathcal{D}}) - D(\hat{p}_X^{\mathcal{D}} \| p_X^\theta). \end{aligned}$$

6.2 Warm-up: MLE in Bayesian networks

It can be easier to start with directed models, because the normalization is local.

Bayesian network factorization. Let G be a DAG on nodes $[n]$. Write $\text{pa}(i)$ for the parent set of node i . A Bayesian network specifies the joint distribution as

$$p_X(x) = \prod_{i=1}^n p_{X_i|X_{\text{pa}(i)}}(x_i | x_{\text{pa}(i)}).$$

For discrete variables, each conditional can be defined by a table of probabilities. We parameterize the local conditional p_{θ_i} by non-negative real numbers $\{\theta_i(x_i | x_{\text{pa}(i)})\}$ for $x_i \in \mathcal{X}_i$ and $x_{\text{pa}(i)} \in \mathcal{X}_{\text{pa}(i)}$. These unnormalized numbers represent the conditional distribution over x_i given $x_{\text{pa}(i)}$ via

$$p_{X_i|X_{\text{pa}(i)}}^{\theta_i}(x_i | x_{\text{pa}(i)}) = \frac{\theta_i(x_i | x_{\text{pa}(i)})}{\sum_{x' \in \mathcal{X}_i} \theta_i(x' | x_{\text{pa}(i)})}.$$

Log-likelihood decomposes into local terms. Using this, the normalized log-likelihood of the dataset can be written as

$$\begin{aligned} \frac{1}{N} \mathcal{L}(\theta) &= \frac{1}{N} \ln \prod_{\ell=1}^N \prod_{i=1}^n \frac{\theta_i(x_i^{(\ell)} | x_{\text{pa}(i)}^{(\ell)})}{\sum_{x' \in \mathcal{X}_i} \theta_i(x' | x_{\text{pa}(i)}^{(\ell)})} \\ &= \frac{1}{N} \sum_{\ell=1}^N \sum_{i=1}^n \ln \frac{\theta_i(x_i^{(\ell)} | x_{\text{pa}(i)}^{(\ell)})}{\sum_{x' \in \mathcal{X}_i} \theta_i(x' | x_{\text{pa}(i)}^{(\ell)})} \\ &= \widehat{\mathbb{E}}_{\mathcal{D}} \left[\sum_{i=1}^n \ln \frac{\theta_i(x_i^{(\ell)} | x_{\text{pa}(i)}^{(\ell)})}{\sum_{x' \in \mathcal{X}_i} \theta_i(x' | x_{\text{pa}(i)}^{(\ell)})} \right] \\ &= \sum_{i=1}^n \widehat{\mathbb{E}}_{\mathcal{D}} \left[\ln \frac{\theta_i(x_i^{(\ell)} | x_{\text{pa}(i)}^{(\ell)})}{\sum_{x' \in \mathcal{X}_i} \theta_i(x' | x_{\text{pa}(i)}^{(\ell)})} \right] \\ &= \sum_{i=1}^n \widehat{\mathbb{E}}_{\mathcal{D}} \left[\ln \frac{\theta_i(x_i^{(\ell)} | x_{\text{pa}(i)}^{(\ell)})}{\sum_{x' \in \mathcal{X}_i} \theta_i(x' | x_{\text{pa}(i)}^{(\ell)})} \cdot \frac{\widehat{p}_{X_i|X_{\text{pa}(i)}}^{\mathcal{D}}(x_i | x_{\text{pa}(i)}^{(\ell)})}{\widehat{p}_{X_i|X_{\text{pa}(i)}}^{\mathcal{D}}(x_i | x_{\text{pa}(i)}^{(\ell)})} \right] \\ &= - \sum_{i=1}^n \widehat{\mathbb{E}}_{\mathcal{D}} \left[\ln \frac{\widehat{p}_{X_i|X_{\text{pa}(i)}}^{\mathcal{D}}(x_i^{(\ell)} | x_{\text{pa}(i)}^{(\ell)})}{\theta_i(x_i^{(\ell)} | x_{\text{pa}(i)}^{(\ell)})} \right] - \sum_{i=1}^n H(\widehat{p}_{X_i|X_{\text{pa}(i)}}^{\mathcal{D}}) - \sum_{i=1}^n \widehat{\mathbb{E}}_{\mathcal{D}} \left[\ln \sum_{x' \in \mathcal{X}_i} \theta_i(x' | x_{\text{pa}(i)}^{(\ell)}) \right] \\ &= - \sum_{i=1}^n D(\widehat{p}_{X_i|X_{\text{pa}(i)}}^{\mathcal{D}} \| p_{X_i|X_{\text{pa}(i)}}^{\theta_i}) - \sum_{i=1}^n H(\widehat{p}_{X_i|X_{\text{pa}(i)}}^{\mathcal{D}}) - \sum_{i=1}^n \widehat{\mathbb{E}}_{\mathcal{D}} \left[\ln \sum_{x' \in \mathcal{X}_i} \theta_i(x' | x_{\text{pa}(i)}^{(\ell)}) \right]. \end{aligned}$$

The sum of KL divergences in the first term is independent of the θ_i normalizations and minimized if and only if $\widehat{p}_{X_i|X_{\text{pa}(i)}}^{\mathcal{D}} = p_{X_i|X_{\text{pa}(i)}}^{\theta_i}$ for all $i \in [n]$. The second term is independent of θ . The last term is zero if we choose all the θ_i parameters to define normalized distributions.

Closed-form MLE for conditional probability tables. Fix a node i and a parent configuration $u \in \mathcal{X}_{\text{pa}(i)}$. Define parent-child configuration counts

$$N_i(x_i, x_{\text{pa}(i)}) := \sum_{\ell=1}^N \mathbb{I}\{x_i^{(\ell)} = x_i, x_{\text{pa}(i)}^{(\ell)} = x_{\text{pa}(i)}\},$$

This implies that

$$\hat{\theta}_i^{\text{ML}}(x_i | u) = \frac{N_i(x_i, u)}{\sum_{x \in \mathcal{X}_i} N_i(x, u)}.$$

So the MLE rule of thumb is: *normalize the empirical conditional frequencies*.

Remark 2 (Why this is easy). Bayesian networks have built-in local normalization: each factor is a conditional distribution, so there is no global partition function. As a result, the MLE is just counting and normalizing.

Smoothing and zero counts. If $N_i(u) = 0$ for some parent configuration u , the MLE is undefined. Even when $N_i(u)$ is small, the MLE can overfit. A common fix is Laplace/Dirichlet smoothing:

$$\hat{\theta}_i^{\text{Dir}(\alpha)}(x_i | u) = \frac{N_i(x_i, u) + \alpha}{N_i(u) + \alpha|\mathcal{X}_i|},$$

which corresponds to a symmetric Dirichlet prior with pseudo-count α . We will not emphasize Bayesian learning today, but it is worth remembering this practical issue.

6.3 Factor graphs: learning factor node functions

We now switch to the factor-graph viewpoint. Let $G = (V, F, E)$ be a factor graph with variable nodes $i \in V = [n]$ and factor nodes $a \in F$. In contrast to earlier descriptions, this note does not require each variable to use the same alphabet. For each factor node a , let $\partial a \subseteq V$ denote its neighboring variable indices, and write

$$x_{\partial a} := (x_i)_{i \in \partial a} \in \mathcal{X}_{\partial a} := \prod_{i \in \partial a} \mathcal{X}_i.$$

6.3.1 Energy-factor notation and exponential-family parameterization

We will write the probability of a configuration $x \in \mathcal{X}$ under the assumed model as a product of *factor functions*

$$p_X^\theta(x) = \frac{1}{Z(\theta)} \prod_{a \in F} f_a(x_{\partial a}; \theta_a), \quad Z(\theta) := \sum_{x \in \mathcal{X}} \prod_{a \in F} f_a(x_{\partial a}; \theta_a).$$

The quantity $Z(\theta)$ is the *partition function* and is the source of most computational difficulty in undirected models.

Remark 3. Directed models such as Bayesian networks can also be written in factor-graph form. In that case, learning is simplified by the fact that $Z(\theta) = 1$ identically.

Local energies. We define the local energy associated with factor a by

$$E_a(x_{\partial a}; \theta_a) := -\ln f_a(x_{\partial a}; \theta_a).$$

Using this, the global energy is

$$E_\theta(x) := \sum_{a \in F} E_a(x_{\partial a}; \theta_a),$$

and the model takes the standard Gibbs / energy-based form

$$p_X^\theta(x) = \frac{1}{Z(\theta)} \exp(-E_\theta(x)), \quad Z(\theta) = \sum_{x \in \mathcal{X}} \exp(-E_\theta(x)).$$

Exponential-family factors. An exponential family of distributions is defined by a set of parameters and sufficient statistics for those parameters. In this lecture, we only deal with the simple case of unrestricted discrete distributions where the parameters are the negative log probabilities of each outcome and a sufficient statistic for a dataset is its histogram.

Definition 4. A finite set $\mathcal{S}$ set with $m = |\mathcal{S}|$ is ordered if it has a bijection $\sigma: \mathcal{S} \rightarrow [m]$ that maps $s \in \mathcal{S}$ to its rank in the set. With this, one can define the *indicator vector* (or one-hot encoding) of $s \in \mathcal{S}$ via $T: \mathcal{S} \rightarrow \mathbb{R}^m$ where $[T(s)]_{\sigma(s)} = 1$ and $[T(s)]_j = 0$ for all $j \neq \sigma(s)$. We also extend this definition to any finite set by choosing a fixed (but arbitrary) ordering for that set.

For the case of a single factor $f(x) = e^{-E(x;\theta)}$ with $x \in \mathcal{X}$ and parameter $\theta \in \mathbb{R}^{|\mathcal{X}|}$, we can define the sufficient statistic $T: \mathcal{X} \rightarrow \mathbb{R}^{|\mathcal{X}|}$ to be the one-hot encoding of $x \in \mathcal{X}$. In particular, if $\mathcal{X} = \{00, 01, 10, 11\}$, then $T(00) = (1, 0, 0, 0)$, $T(01) = (0, 1, 0, 0)$, etc. Then, we can write $E(x; \theta) = -\theta^\top T(x)$ and $f(x; \theta) = \exp(-E(x; \theta)) = \exp(\theta^\top T(x))$.

We assume each factor lives in the canonical exponential family described above. For each $a \in F$, fix a sufficient-statistic function $T_a: \mathcal{X}_{\partial a} \rightarrow \mathbb{R}^{|\mathcal{X}_{\partial a}|}$ that maps each configuration to a vector. Then, we define parameters $\theta_a \in \mathbb{R}^{|\mathcal{X}_{\partial a}|}$ and

$$f_a(x_{\partial a}; \theta_a) = \exp(\theta_a^\top T_a(x_{\partial a})).$$

With this convention, one gets the energies

$$E_a(x_{\partial a}; \theta_a) = -\theta_a^\top T_a(x_{\partial a}).$$

It is sometimes preferred to use energies that look like positive “costs”. To do this, one can simply flip the sign by redefining T_a with a sign flip.

Remark 5 (Indicator statistics as table parameters). If $x_{\partial a}$ is discrete and you want a fully general factor, then take the indicator statistics: for each $u \in \mathcal{X}_{\partial a}$ define the indicator vector $[T_a(x_{\partial a})]_u = \mathbb{I}\{x_{\partial a} = u\}$. Then $f_a(x_{\partial a}) = \exp([\theta_a]_{x_{\partial a}})$ is an arbitrary positive table. There is also one extra degree of freedom because adding a constant to all $[\theta_a]_u$ multiplies f_a by a constant and is absorbed into $Z(\theta)$. A common fix is $\sum_u [\theta_a]_u = 0$ or pinning one entry (e.g. $[\theta_a]_{u_0} = 0$).

Using this setup, we can write the model as

$$p_X^\theta(x) = \frac{1}{Z(\theta)} \exp\left(\sum_{b \in F} \theta_b^\top T_b(x_{\partial b})\right), \quad Z(\theta) = \sum_{x \in \mathcal{X}} \exp\left(\sum_{b \in F} \theta_b^\top T_b(x_{\partial b})\right). \quad (6.1)$$

Lemma 6 (Gradient of the log-partition function). *For every $a \in F$, it follows from (6.1) that*

$$\nabla_{\theta_a} \ln Z(\theta) = \mathbb{E}_{p_X^\theta}[T_a(X_{\partial a})].$$

Proof. Since $\mathcal{X}$ is finite, we may differentiate under the summation sign:

$$\begin{aligned} \nabla_{\theta_a} \ln Z(\theta) &= \frac{1}{Z(\theta)} \nabla_{\theta_a} Z(\theta) \\ &= \frac{1}{Z(\theta)} \sum_{x \in \mathcal{X}} \nabla_{\theta_a} \exp\left(\sum_{b \in F} \theta_b^\top T_b(x_{\partial b})\right) \\ &= \frac{1}{Z(\theta)} \sum_{x \in \mathcal{X}} \exp\left(\sum_{b \in F} \theta_b^\top T_b(x_{\partial b})\right) T_a(x_{\partial a}) \\ &= \sum_{x \in \mathcal{X}} p_X^\theta(x) T_a(x_{\partial a}) \\ &= \mathbb{E}_{p_X^\theta}[T_a(X_{\partial a})]. \end{aligned}$$

□

6.3.2 Log-likelihood for fully observed data

Given an i.i.d. dataset $\mathcal{D} = \{x^{(1)}, \dots, x^{(N)}\}$ with full observations, the log-likelihood is

$$\begin{aligned}\mathcal{L}(\theta) &:= \ln p_X^\theta(\mathcal{D}) = \sum_{\ell=1}^N \ln p_X^\theta(x^{(\ell)}) \\ &= \sum_{\ell=1}^N \left[\sum_{b \in F} \ln f_b(x_{\partial b}^{(\ell)}; \theta_b) - \ln Z(\theta) \right] \\ &= \sum_{\ell=1}^N \left[\sum_{b \in F} \theta_b^\top T_b(x_{\partial b}^{(\ell)}) - \ln Z(\theta) \right].\end{aligned}$$

Equivalently, dividing by N and using empirical averages,

$$\frac{1}{N} \mathcal{L}(\theta) = \sum_{b \in F} \theta_b^\top \widehat{\mathbb{E}}_{\mathcal{D}}[T_b(X_{\partial b})] - \ln Z(\theta).$$

The first term is easy to compute from the data. In particular, since T returns a one-hot vector, the expectation term equals the empirical distribution vector

$$\widehat{\mathbb{E}}_{\mathcal{D}}[T_b(X_{\partial b})] = (p_{X_{\partial b}}^{\mathcal{D}}(x_{\partial b}))_{x_{\partial b} \in \mathcal{X}_{\partial b}}.$$

The second term is independent of the data and depends on θ through the partition function.

Gradient: data moments minus model moments Differentiate with respect to θ_a :

$$\nabla_{\theta_a} \left(\frac{1}{N} \mathcal{L}(\theta) \right) = \widehat{\mathbb{E}}_{\mathcal{D}}[T_a(X_{\partial a})] - \nabla_{\theta_a} \ln Z(\theta).$$

From Lemma 6, we get the standard derivative identity for exponential-family energy models:

$$\nabla_{\theta_a} \ln Z(\theta) = \mathbb{E}_{p_X^\theta}[T_a(X_{\partial a})].$$

Therefore,

$$\nabla_{\theta_a} \left(\frac{1}{N} \mathcal{L}(\theta) \right) = \widehat{\mathbb{E}}_{\mathcal{D}}[T_a(X_{\partial a})] - \mathbb{E}_{p_X^\theta}[T_a(X_{\partial a})].$$

This is a standard learning approach:

update parameters to match model sufficient statistics to empirical sufficient statistics.

Proposition 7 (Moment matching at the MLE (interior optimum)). *If $\hat{\theta}$ maximizes $\mathcal{L}(\theta)$ in the interior of the parameter space, then for every factor a ,*

$$\mathbb{E}_{p_X^{\hat{\theta}}}[T_a(X_{\partial a})] = \widehat{\mathbb{E}}_{\mathcal{D}}[T_a(X_{\partial a})].$$

Inference is the bottleneck. The empirical averages are trivial, but the model expectations $\mathbb{E}_{p_X^\theta}[T_a(X_{\partial a})]$ require computing the marginal distribution of $X_{\partial a}$ under p_X^θ . This is an inference problem. Thus, in undirected / factor-graph models, *learning is an outer loop wrapped around repeated inference.*

6.3.3 Learning on trees: exact BP in the inner loop

If the factor graph is a tree, sum-product BP computes exact factor marginals $p_{X_{\partial a}}^\theta(x_{\partial a})$ and thus

$$\mathbb{E}_{p_X^\theta}[T_a(X_{\partial a})] = \sum_{x_{\partial a}} p_{X_{\partial a}}^\theta(x_{\partial a}) T_a(x_{\partial a})$$

exactly and efficiently.

A simple gradient-ascent scheme is therefore:

$$\theta_a^{(t+1)} = \theta_a^{(t)} + \eta_t \left(\widehat{\mathbb{E}}_{\mathcal{D}}[T_a(X_{\partial a})] - \mathbb{E}_{p_X^{\theta^{(t)}}}[T_a(X_{\partial a})] \right),$$

where $\eta_t > 0$ is a step size. Each iteration uses BP to compute the current model marginals.

Remark 8. Two implementation details worth keeping in mind are: (i) on trees, BP recovers $\ln Z(\theta)$ (along with marginals), and (ii) some model classes with local normalization constraints admit closed-form M-step style updates instead of generic gradient steps.

6.3.4 Learning on loopy graphs: approximate inference in the inner loop

When the factor graph has cycles, BP is no longer guaranteed to be exact or even to converge. Nevertheless the same gradient identity holds. Typical approximations include:

1. **Loopy BP gradients:** run loopy BP to get approximate factor beliefs $b_a(x_{\partial a})$, and plug them into $\mathbb{E}_{p_X^\theta}[T_a] \approx \sum_{x_{\partial a}} b_a(x_{\partial a}) T_a(x_{\partial a})$.
2. **Sampling-based estimates:** use MCMC (e.g. Gibbs sampling) to approximate $\mathbb{E}_{p_X^\theta}[T_a]$ by Monte Carlo.
3. **Pseudo-likelihood:** replace the likelihood by a product of local conditional likelihoods to avoid $Z(\theta)$.
4. **Contrastive divergence / persistent chains.** Approximate the model expectation by running a small number of MCMC steps from data. This is historically popular for energy-based models.

In all cases, the message is the same: *learning is easy once inference is easy*.

6.4 Missing data / latent variables: EM with BP

We now move to the setting where some variables are unobserved. Let $X = (Y, H)$ where Y are observed variables and H are hidden (latent) variables. Each sample provides only $y^{(\ell)}$. We wish to maximize the observed-data log-likelihood

$$\mathcal{L}_{\text{obs}}(\theta) := \sum_{\ell=1}^N \ln p_Y^\theta(y^{(\ell)}), \quad p_Y^\theta(y) = \sum_h p_X^\theta(y, h).$$

The difficulty is the log-sum over hidden states.

6.4.1 EM lower bound and the Q -function

EM is an alternating maximization procedure. For each data point $y^{(\ell)}$, introduce an auxiliary distribution $q^{(\ell)}(h)$ over hidden states. Using Jensen's inequality:

$$\ln \sum_h p_X^\theta(y, h) = \ln \sum_h q(h) \frac{p_X^\theta(y, h)}{q(h)} \geq \sum_h q(h) \ln \frac{p_X^\theta(y, h)}{q(h)}.$$

Summing over ℓ ,

$$\mathcal{L}_{\text{obs}}(\theta) \geq \sum_{\ell=1}^N \sum_h q^{(\ell)}(h) \ln p_X^\theta(y^{(\ell)}, h) - \sum_{\ell=1}^N \sum_h q^{(\ell)}(h) \ln q^{(\ell)}(h).$$

Define the EM functional

$$\mathcal{F}(\theta, \{q^{(\ell)}\}) := \sum_{\ell=1}^N \mathbb{E}_{q^{(\ell)}} [\ln p_X^\theta(Y = y^{(\ell)}, H)] + \sum_{\ell=1}^N H(q^{(\ell)}),$$

where $H(q) = -\sum_h q(h) \ln q(h)$.

Remark 9 (What EM does?). The EM algorithm alternates:

- E-step: choose $q^{(\ell)}$ to maximize $\mathcal{F}$ for fixed θ ;
- M-step: choose θ to maximize $\mathcal{F}$ for fixed $\{q^{(\ell)}\}$.

Each step does is guaranteed to maintain or decrease the observed-data log-likelihood.

E-step: posterior inference with evidence. In a factor graph, conditioning on $Y = y^{(\ell)}$ is implemented by clamping the observed variable nodes (or equivalently adding unary “evidence factors” that zero out inconsistent states). Then the E-step requires computing local posteriors such as

$$p_{X_{\partial a}|Y}^{\theta^{\text{old}}}(x_{\partial a} | y^{(\ell)}),$$

which on trees are obtained exactly by BP.

In terms of sufficient statistics, the E-step provides *soft* (posterior-averaged) statistics:

$$\hat{T}_{a,\text{soft}} := \frac{1}{N} \sum_{\ell=1}^N \mathbb{E}_{p_{H|Y}^{\theta^{\text{old}}}(h|y^{(\ell)})} [T_a(X_{\partial a})] = \frac{1}{N} \sum_{\ell=1}^N \sum_{x_{\partial a}} p_{X_{\partial a}|Y}^{\theta^{\text{old}}}(x_{\partial a} | y^{(\ell)}) T_a(x_{\partial a}).$$

M-step: maximize expected complete-data log-likelihood. Using the exponential-family energy form, we get

$$\ln p_X^\theta(x) = \sum_{a \in F} \theta_a^\top T_a(x_{\partial a}) - \ln Z(\theta).$$

So the M-step objective becomes

$$Q(\theta | \theta^{\text{old}}) = \sum_{\ell=1}^N \left(\theta^\top \mathbb{E}_{p_{H|Y}^{\theta^{\text{old}}}(h|y^{(\ell)})} [T(y^{(\ell)}, h)] - \ln Z(\theta) \right).$$

Thus, compared to the fully observed case, the empirical features are replaced by *expected* features under the posterior and we have

$$\frac{1}{N} Q(\theta | \theta^{\text{old}}) = \sum_{a \in F} \theta_a^\top \hat{T}_{a,\text{soft}} - \ln Z(\theta).$$

Therefore the M-step has the same structure as the fully observed case, but with empirical statistics replaced by soft statistics. Differentiating with θ gives

$$\nabla_{\theta_a} \left(\frac{1}{N} Q(\theta \mid \theta^{\text{old}}) \right) = \hat{T}_{a,\text{soft}} - \mathbb{E}_{p_X^\theta} [T_a(X_{\partial a})].$$

So again we have “(soft) data moments minus model moments.”

Two inference problems. Notice what EM requires:

- In the E-step: inference in the *clamped* model $p_{X|Y}^{\theta^{\text{old}}}(\cdot \mid y^{(\ell)})$ to compute soft statistics.
- In the M-step (for undirected models): inference in the *unclamped* model p_X^θ to compute $\mathbb{E}_{p_X^\theta} [T_a]$ (unless you adopt an M-step approximation or a locally normalized parameterization).

On trees both are exact via BP; on loopy graphs both are typically approximate (loopy BP, sampling, etc.).

6.4.2 EM gradient form and “soft” moment matching

Define the *soft empirical feature average*

$$\hat{f}_{\text{soft}} := \frac{1}{N} \sum_{\ell=1}^N \mathbb{E}_{p_{H|Y}^{\theta^{\text{old}}}(h|y^{(\ell)})} [f(y^{(\ell)}, h)].$$

Then the gradient of $\frac{1}{N} Q(\theta \mid \theta^{\text{old}})$ is

$$\nabla_{\theta} \left(\frac{1}{N} Q(\theta \mid \theta^{\text{old}}) \right) = \hat{f}_{\text{soft}} - \mathbb{E}_{p_X^\theta} [f(X)].$$

So it is the same “data minus model” structure as before, except that “data” is replaced by posterior expectations from the E-step.

Where BP appears. In many discrete models, $\hat{f}_{\text{soft}}$ can be computed from local marginals $p_{X_{\partial a}|Y}^{\theta^{\text{old}}}(x_{\partial a} \mid y^{(\ell)})$, which are exactly what BP estimates (or computes exactly on trees).

6.4.3 EM + BP: algorithm template

When the factor graph is a tree, an EM iteration can be written as:

1. **E-step (BP with evidence).** For each $\ell = 1, \dots, N$: clamp $Y = y^{(\ell)}$ and run BP to obtain exact marginals $p_{X_{\partial a}|Y}^{\theta^{\text{old}}}(x_{\partial a} \mid y^{(\ell)})$ for each factor a . Accumulate soft sufficient statistics $\hat{T}_{a,\text{soft}}$.
2. **M-step (optimize Q).** In the standard EM algorithm, the M-step solves for a θ with zero derivative in order to find a maximum over θ for the current q . Instead, modern approaches often update θ by (e.g.) gradient ascent:

$$\theta_a \leftarrow \theta_a + \eta \left(\hat{T}_{a,\text{soft}} - \mathbb{E}_{p_X^\theta} [T_a(X_{\partial a})] \right),$$

where $\mathbb{E}_{p_X^\theta} [T_a]$ is computed by a BP run on the unclamped tree at the current θ . (Depending on the model class, some M-steps admit closed forms; for general undirected exponential families, iterative optimization is typical.)

Rule of thumb: “compute soft counts then normalize” In many familiar latent-variable models (mixtures, HMMs, etc.), the M-step reduces to:

estimate posterior (soft) counts and use them like hard counts to estimate fully observed MLE.

On chains (HMMs), BP is forward-backward; on trees, BP is sum-product; on loopy graphs, EM typically becomes “approximate EM” because the E-step posteriors are approximate.

6.5 Example: EM for a discrete mixture model

Consider a two-variable Bayesian network with a latent class variable $C \in [K]$ and an observed variable $Y \in \mathcal{Y}$. The model is

$$p_{C,Y}^\theta(c, y) = p_C^\theta(c) p_{Y|C}^\theta(y | c) = \pi_c \phi_{y|c},$$

where the parameters $\theta = (\pi, \phi)$ satisfy

$$\pi_c \geq 0, \quad \sum_{c=1}^K \pi_c = 1, \quad \phi_{y|c} \geq 0, \quad \sum_{y \in \mathcal{Y}} \phi_{y|c} = 1 \quad \forall c.$$

Given observed data $\mathcal{D} = \{y^{(1)}, \dots, y^{(N)}\}$ (classes unobserved), the observed-data log-likelihood is

$$\mathcal{L}_{\text{obs}}(\theta) = \sum_{\ell=1}^N \ln p_Y^\theta(y^{(\ell)}) = \sum_{\ell=1}^N \ln \left(\sum_{c=1}^K \pi_c \phi_{y^{(\ell)}|c} \right).$$

E-step (class posteriors). For current parameters $\theta^{\text{old}} = (\pi^{\text{old}}, \phi^{\text{old}})$, compute the soft class assignments

$$\gamma_{\ell c} := p_{C|Y}^{\theta^{\text{old}}}(c | y^{(\ell)}) = \frac{\pi_c^{\text{old}} \phi_{y^{(\ell)}|c}^{\text{old}}}{\sum_{j=1}^K \pi_j^{\text{old}} \phi_{y^{(\ell)}|j}^{\text{old}}}.$$

M-step (maximize expected complete log-likelihood). The EM objective for fixed $\gamma_{\ell c}$ is

$$Q(\theta | \theta^{\text{old}}) = \sum_{\ell=1}^N \sum_{c=1}^K \gamma_{\ell c} (\ln \pi_c + \ln \phi_{y^{(\ell)}|c}),$$

subject to normalization constraints on π and each conditional table $\phi_{\cdot|c}$. Lagrange multipliers give closed-form updates:

$$\pi_c^{\text{new}} = \frac{1}{N} \sum_{\ell=1}^N \gamma_{\ell c}, \quad \phi_{y|c}^{\text{new}} = \frac{\sum_{\ell=1}^N \gamma_{\ell c} \mathbb{I}\{y^{(\ell)} = y\}}{\sum_{\ell=1}^N \gamma_{\ell c}}.$$

EM algorithm (mixture model).

1. Initialize π to have full support and ϕ to have full support in each class.
2. E-step: compute $\gamma_{\ell c}$ for all ℓ, c .
3. M-step: update $\pi_c, \phi_{y|c}$ using the formulas above.
4. Repeat E/M steps until the increase in $\mathcal{L}_{\text{obs}}(\theta)$ is below a tolerance.

This is exactly the recipe above: E-step computes posterior expectations (soft counts), and M-step normalizes those soft counts to update local CPT parameters.

6.6 Example: EM on a hidden Markov model (HMM)

Although HMMs are usually introduced as directed models, they have a clean factor graph view and provide an excellent showcase for BP/EM. Let $S = (S_1, \dots, S_T) \in \mathcal{S}^T$ be hidden states and $(Y_1, \dots, Y_T) \in \mathcal{Y}^T$ be observed symbols. The joint distribution factors as

$$p_{S,Y}^\theta(s, y) = p_{S_1}^\theta(s_1) \prod_{t=2}^T p_{S_t|S_{t-1}}^\theta(s_t | s_{t-1}) \prod_{t=1}^T p_{Y_t|S_t}^\theta(y_t | s_t).$$

This is a chain factor graph. BP on a chain is exact and known as the forward-backward algorithm.

E-step. Compute posteriors:

$$\gamma_t(i) := p_{S_t|Y}^{\theta^{\text{old}}}(i|y), \quad \xi_t(i, j) := p_{S_{t-1}, S_t|Y}^{\theta^{\text{old}}}(i, j|y).$$

These are obtained from BP messages.

M-step. Update transition probabilities (via θ) by normalized expected counts:

$$p_{S_t|S_{t-1}}^\theta(j|i) = \frac{\sum_{t=2}^T \xi_t(i, j)}{\sum_{t=2}^T \sum_{j'} \xi_t(i, j')}.$$

Update emission probabilities (via θ) similarly:

$$p_{Y_t|S_t}^\theta(y|i) = \frac{\sum_{t=1}^T \gamma_t(i) \mathbb{I}\{y_t = y\}}{\sum_{t=1}^T \gamma_t(i)}.$$

This is a concrete template: compute local marginals by BP, then normalize. If the HMM is known to be time-invariant, then the E-step counts can be pooled across time steps to get better estimates.

6.7 Summary: what to remember

1. **Bayesian networks (known DAG, fully observed):** MLE is counting + normalization (closed form).
2. **Undirected/factor graph models (fully observed):**

$$\nabla_\theta \left(\frac{1}{N} \ln p_X^\theta(\mathcal{D}) \right) = \widehat{\mathbb{E}}_{\mathcal{D}}[T(X)] - \mathbb{E}_{p_X^\theta}[T(X)],$$

where $T(x)$ is a concatenation of $T_a(x_{\partial a})$ over all factors a . Learning is outer-loop optimization while inference (and computing marginals) is inner-loop computation.

3. **Missing data / latent variables: EM.** E-step runs inference to compute posterior expectations of sufficient statistics. M-step updates parameters using these “soft” statistics (sometimes closed form, sometimes requiring optimization). BP is the natural inference engine in factor graphs (exact on trees, approximate on loopy graphs).

6.8 Exercises

Exercise 1 (EM soft statistics). *Derive the expression for $\hat{T}_{a,\text{soft}}$ in the E-step and explain why it can be computed from the factor marginals $p_{X_{\partial a}|Y}^{\theta_{\text{old}}}(x_{\partial a} | y^{(\ell)})$ returned by BP on a tree.*

Exercise 2 (Gauge freedom for table factors). *Let $\mathcal{X}_{\partial a}$ be finite and define $f_a(u) = \exp(\theta_{a,u})$. Show that adding a constant c to all $\theta_{a,u}$ multiplies f_a by e^c and does not change p_X^θ after renormalization. Give one gauge constraint that removes this redundancy.*

Exercise 3 (MLE for a single CPT via direct optimization). *Let $X \in \{1, \dots, r\}$ and $U \in \{1, \dots, s\}$, and suppose the conditional distribution is a CPT $p_\theta(x | u) = \theta_{x|u}$ with constraints $\sum_{x=1}^r \theta_{x|u} = 1$ for each u . Given an i.i.d. dataset $\mathcal{D} = \{(x^{(\ell)}, u^{(\ell)})\}_{\ell=1}^N$, define counts $N(x, u) = \sum_{\ell=1}^N \mathbb{I}\{x^{(\ell)} = x, u^{(\ell)} = u\}$ and $N(u) = \sum_x N(x, u)$.*

1. *Write the (normalized) log-likelihood $\frac{1}{N}\mathcal{L}(\theta)$ in terms of $N(x, u)$ and $\theta_{x|u}$.*
2. *Using Lagrange multipliers (one constraint per u), derive the MLE for $\theta_{x|u}$.*
3. *Explain what happens if $N(u) = 0$ and give a simple practical fix.*

Exercise 4 (Convexity/concavity of the undirected log-likelihood). *Consider the exponential-family factor graph model*

$$p^\theta(x) = \frac{1}{Z(\theta)} \exp\left(\sum_{a \in F} \theta_a^\top T_a(x_{\partial a})\right), \quad \theta = (\theta_a)_{a \in F}.$$

Let $\frac{1}{N}\mathcal{L}(\theta)$ be the normalized log-likelihood for fully observed data.

1. *Show that $\nabla_\theta \ln Z(\theta) = \mathbb{E}_{p^\theta}[T(X)]$, where $T(X)$ is the concatenation of all $T_a(X_{\partial a})$.*
2. *Show that the Hessian satisfies $\nabla_\theta^2 \ln Z(\theta) = \text{Cov}_{p^\theta}(T(X))$.*
3. *Conclude that $\ln Z(\theta)$ is convex and that $\ell(\theta)$ is concave.*

Exercise 5 (One-step contrastive divergence update (CD-1) for a log-linear model). *Consider a discrete energy-based model*

$$p^\theta(x) = \frac{1}{Z(\theta)} \exp(\theta^\top T(x)).$$

Let $\hat{\mathbb{E}}_{\mathcal{D}}[T(X)]$ be the empirical feature average. In CD-1, one approximates the model expectation $\mathbb{E}_{p^\theta}[T(X)]$ by sampling a data point $X^0 \sim \hat{p}_X^{\mathcal{D}}$, performing a Gibbs step to get X^1 , and using $T(X^1)$.

1. *Write the exact gradient of the normalized log-likelihood $\frac{1}{N}\mathcal{L}(\theta)$.*
2. *Write the CD-1 approximate gradient $\hat{g}_{\text{CD1}}(\theta)$ in terms of $T(X^0)$ and $T(X^1)$.*
3. *Give the resulting CD-1 update rule with step size η .*

Exercise 6 (EM monotonicity via a KL decomposition). *Recall the EM functional*

$$\mathcal{F}(\theta, \{q^{(\ell)}\}) = \sum_{\ell=1}^N \mathbb{E}_{q^{(\ell)}}[\ln p^\theta(y^{(\ell)}, H)] + \sum_{\ell=1}^N H(q^{(\ell)}).$$

1. *Show that for each ℓ ,*

$$\ln p^\theta(y^{(\ell)}) = \mathcal{F}_\ell(\theta, q^{(\ell)}) + D(q^{(\ell)}(\cdot) \| p^\theta(H | y^{(\ell)})),$$

where $\mathcal{F}_\ell$ is the ℓ th summand of $\mathcal{F}$.

2. Use this identity to argue that choosing $q^{(\ell)} = p^{\theta^{\text{old}}}(H \mid y^{(\ell)})$ maximizes $\mathcal{F}$ over q for fixed θ^{old} .
3. Conclude (informally) why the E-step followed by an exact M-step does not decrease the observed-data log-likelihood.

Exercise 7 (HMM forward-backward messages and γ, ξ). Consider an HMM with hidden states $S_t \in \{1, \dots, K\}$ and observations $Y_t \in \mathcal{Y}$ with parameters: initial $\pi(i) = p(S_1 = i)$, transition $A_{ij} = p(S_t = j \mid S_{t-1} = i)$, and emission $B_j(y) = p(Y_t = y \mid S_t = j)$. Given a single observation sequence $y_{1:T}$:

1. Define forward messages $\alpha_t(j)$ and give their recursion (up to normalization).
2. Define backward messages $\beta_t(i)$ and give their recursion (up to normalization).
3. Express $\gamma_t(j) = p(S_t = j \mid y_{1:T})$ in terms of α_t, β_t .
4. Express $\xi_t(i, j) = p(S_{t-1} = i, S_t = j \mid y_{1:T})$ in terms of $\alpha_{t-1}, A, B, \beta_t$.

Exercise 8 (EM M-step for an HMM with multiple sequences). Suppose we observe N independent sequences $\{y_{1:T_\ell}^{(\ell)}\}_{\ell=1}^N$ from the same time-homogeneous HMM parameters (π, A, B) . Let $\gamma_t^{(\ell)}(j)$ and $\xi_t^{(\ell)}(i, j)$ be the posteriors computed in the E-step for sequence ℓ .

1. Derive the M-step update for the initial distribution $\pi(j)$.
2. Derive the M-step update for the transition probabilities A_{ij} .
3. Assume emissions are categorical over $\mathcal{Y}$ (finite alphabet). Derive the M-step update for $B_j(y)$.